Note: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

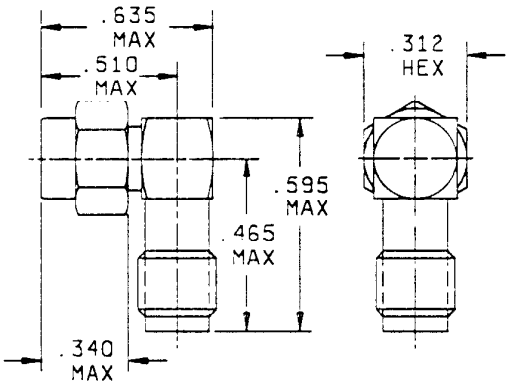
MIL-PRF-55339/2A(USAF)
4 August 1975
SUPERSEDING
MIL-A-55339/2
1 August 1972

PERFORMANCE SPECIFICATION

ADAPTERS, CONNECTOR, COAXIAL, RADIO FREQUENCY, (SERIES SMA), CLASS 2, RIGHT ANGLE, MALE TO FEMALE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapters described herein shall consist of this document and the latest issue of Specification MIL-PRF-55339.



INCHES	MM
.312	7.92
.340	8.64
.465	11.81
.510	12.95
.595	15.11
.635	16/13

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Wrench flats to accommodate standard wrench per H-28 appendix 10.

DESIGN AND CONSTRUCTION:

General configuration	See figure 1.
Center contact spring member	Copper beryllium.
Impedance	50 ohms, nom.
Working voltage:	
Sea level	335 Vrms.
70,000 feet	85 Vrms.
Frequency range	0 to 12.4 GHz.
Temperature range	-65° to +165° C.

PERFORMANCE (installation torque of 7 to 10 in. lb).

Dimensions - See figures 1 and 2.

Center contact retention:
Axial force - 6 lb, min.
Torque - 4 in. oz, min.

Force to engage and disengage:
See table I.

Coupling proof torque: 15 in. lb, max.

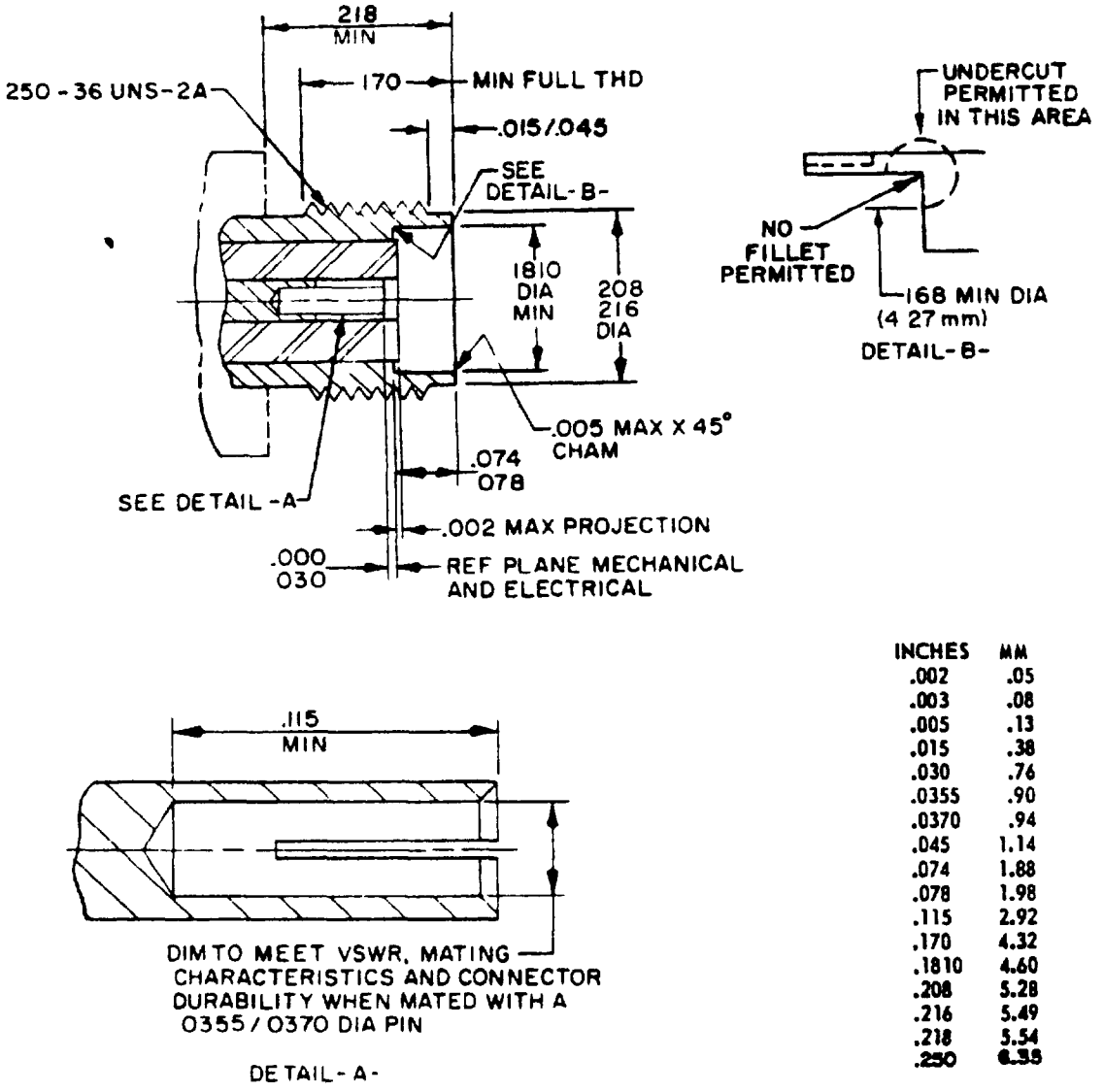
Mating characteristics:
Center contact - See table I.
Outer contact - Not applicable.

Permeability: Less than 2.0.

(A) denotes changes

FSC 5935

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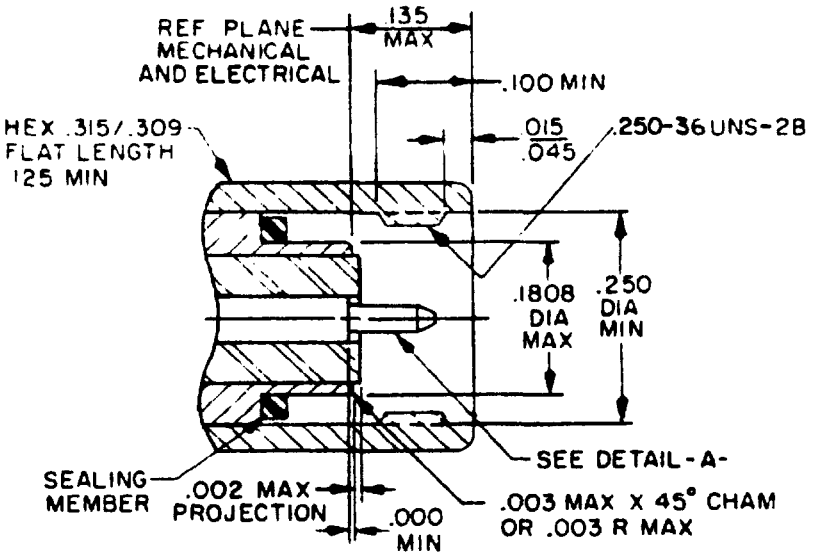
INCHES	MM
.002	.05
.003	.08
.005	.13
.015	.38
.030	.76
.0355	.90
.0370	.94
.045	1.14
.074	1.88
.078	1.98
.115	2.92
.170	4.32
.1810	4.60
.208	5.28
.216	5.49
.218	5.54
.250	6.35

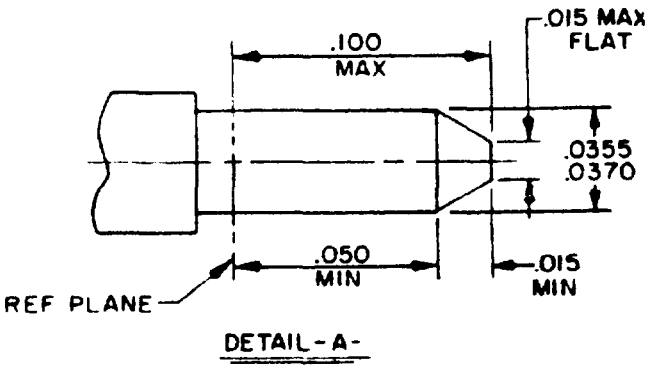
Series SMA female.

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2 Mating dimensions.

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INCHES	MM
.002	.05
.003	.08
.015	.38
.0355	.90
.0370	.94
.045	1.14
.050	1.27
.100	2.54
.125	3.18
.135	3.43
.1808	4.59
.250	6.35
.309	7.85
.315	8.00

Series SMA male.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Mating dimensions - Continued.

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Seal (hermetic pressurized, and weatherproof) Not applicable

Insulation resistance Method 302 of MIL-STD-202, test condition B
5,000 megohms, min

VSWR 1 05 + 010 F, max, at 5 to 12 4 GHz

RF leakage (total) -60 dB, min, at 2 to 3 GHz.

RF insertion loss
(05 $\sqrt{\text{F(GHz)}}$ dB max tested at 6 GHz)

Durability 500 cycles, min, at 12 c/m.

Dielectric withstanding voltage Method 301 of MIL-STD-202
Test voltage - 1,000 Vrms, max, at sea level

Contact resistance (in milliohms, max)

	Initial	After environment
Center contact	4 0	6 0
Outer contact	2 0	N/A

Vibration, high frequency Method 204 of MIL-STD-202, test condition D

Shock (specified pulse) Method 213 of MIL-STD-202, test condition I.

(A) Thermal shock: Method 107 of MIL-STD-202, test condition B.

Moisture resistance Method 106 of MIL-STD-202.
Insulation resistance - 200 megohms, min.

Corona level: 250 V, min
Altitude· 70,000 feet.

RF high potential withstanding voltage
RF voltage - 670 Vrms.
Frequency - 5 MHz.

Salt spray (corrosion) Method 101 of MIL-STD-202, test condition B

Coupling mechanism retention force
60 lbs, min.

(A) MARKING As specified in MIL-A-55339
Part No. M55339/02-30001
-40001.

TABLE I. Performance characteristics

Characteristic	Test value	
	MALE	FEMALE
Force to engage and disengage		
Longitudinal force - - - - - - - - - - - -	N/A	N/A
Torque - - - - - - - - - - - - - - - - - -	2 in lb, max	2 in. lb, max
Mating characteristics (center contact)		
Oversize test pin (inserted 030 045 in deep) - - - - - -	See figure for dim	.0375 +.0001 in. dia
Max test pin (insertion force 3 lb, max) (insertion depth 050/ 075) - - - - - - -		0370 in. dia
Min test pin (withdrawal force 1 oz, min) (insertion depth 050/.075) - - - - - - -		.0355 -.0001 in. dia

Preparing activity·
Air Force - 11

Agent:
DSA - ES

(Project 5935-F995)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street City State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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